

Data Normalization

```
In [1]: from sklearn.preprocessing import MinMaxScaler
import pandas as pd
```

```
In [2]: df_cars = pd.DataFrame([[120000, 11], [250000, 11.5], [175000, 15.8], [350000, 17],
                                columns=['odometer_reading', 'fuel_economy'])
print(df_cars)
```

	odometer_reading	fuel_economy
0	120000	11.0
1	250000	11.5
2	175000	15.8
3	350000	17.0
4	400000	10.0

```
In [3]: def min_max_scaling(df):
        # copy the dataframe
        df_norm = df.copy()
        # apply min-max scaling
        for column in df_norm.columns:
            print(column)
            df_norm[column] = (df_norm[column] - df_norm[column].min()) / (df_norm[column].max() - df_norm[column].min())
            #print( df_norm[column] )
        return df_norm

        # call the min_max_scaling function
        df_cars_normalized = min_max_scaling(df_cars)

        df_cars_normalized
```

	odometer_reading	fuel_economy
0	0.000000	0.142857
1	0.464286	0.214286
2	0.196429	0.828571
3	0.821429	1.000000
4	1.000000	0.000000

```
Out[3]:
```

	odometer_reading	fuel_economy
0	0.000000	0.142857
1	0.464286	0.214286
2	0.196429	0.828571
3	0.821429	1.000000
4	1.000000	0.000000

```
In [4]: # create a scaler object
scaler = MinMaxScaler()
# fit and transform the data
df_norm = pd.DataFrame(scaler.fit_transform(df_cars), columns=df_cars.columns)

df_norm
```

Out[4]:

	odometer_reading	fuel_economy
0	0.000000	0.142857
1	0.464286	0.214286
2	0.196429	0.828571
3	0.821429	1.000000
4	1.000000	0.000000

In [5]:

```
#apply the z-score method in Pandas using the .mean() and .std() methods
def z_score(df):
    # copy the dataframe
    df_std = df.copy()
    # apply the z-score method
    for column in df_std.columns:
        df_std[column] = (df_std[column] - df_std[column].mean()) / df_std[column].std()

    return df_std

# call the z_score function
df_cars_standardized = z_score(df_cars)
df_cars_standardized
```

Out[5]:

	odometer_reading	fuel_economy
0	-1.189512	-0.659120
1	-0.077019	-0.499139
2	-0.718842	0.876693
3	0.778745	1.260647
4	1.206628	-0.979081

In [6]:

```
from sklearn.preprocessing import StandardScaler

# create a scaler object
std_scaler = StandardScaler()
std_scaler

# fit and transform the data
df_std = pd.DataFrame(std_scaler.fit_transform(df_cars), columns=df_cars.columns)

df_std
```

```
Out[6]:
```

	odometer_reading	fuel_economy
0	-1.329915	-0.736918
1	-0.086110	-0.558055
2	-0.803690	0.980173
3	0.870664	1.409446
4	1.349051	-1.094646

Alternatively, we can use the `StandardScaler` class available in the Scikit-learn library to perform the z-score. First, we create a `standard_scaler` object. Then, we calculate the parameters of the transformation (in this case the mean and the standard deviation) using the `.fit()` method. Next, we call the `.transform()` method to apply the standardization to the data frame.

```
In [6]:
```